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Roll No

EC-702-GS

B.E. VII Semester

Examination, December 2020

Grading System (GS)

Satellite Communication

Time : Three Hours

Maximum Marks : 70

Note : i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Discuss various types of antennas used on spacecraft.
b) What is the need of attitude and orbit control? Explain it.
2. Define Kepler's laws of orbiting bodies and derive an equation to show that the third law is true for any orbiting satellite.
3. a) Explain the concept of orbit perturbations. Discuss the principle of inclined orbits.
b) What do you mean by local mean solar point? Discuss about the sun synchronous orbit.
4. a) Discuss the necessity of space segment for our satellite system.
b) What do you mean by cross polarization discrimination? How does it affect the recent satellite systems?
5. a) Draw and discuss elements of basic satellite link showing ground stations and the satellite transponder.

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- b) Explain how intermodulation noise originates in a satellite link and how it can be reduced?
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- 6.
 - a) Derive the expression of Equivalent Isotropic Radiated Power (EIRP) of a space link.
 - b) Determine combine uplink downlink C/N ratio.
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- 7. With the help of diagram explain the working of following:
 - a) Digital DBS TV
 - b) VSAT Earth Station
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- 8. Write short notes on any three of the following:
 - a) Cross polarization discrimination
 - b) Transponders and antenna subsystem
 - c) Satellite radio broad casting
 - d) Interference between satellite circuits
 - e) Frequency allocations for satellite systems
